Encoding Arithmetic In First Order Logic (DRAFT)

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1 Introduction

The purpose of this document is to provide an intermediate representation (IR) for encoding arithmetic (i.e. $+, \times, =, <, >$) using first order logic (FOL) without the equality symbol $=$ or free variables. Furthermore, a first order logic statement of the form $\exists a \forall x \exists y \forall z \forall z'\ P(a, x, y, z, z')$ where P is quantifier free and only contains binary primitive predicates.

2 Overview

We construct a first order sentence (yet to be constructed) in the language of First Order Logic without equality and without bounded variables and without an arbitrary domain where the satisfying model is an IR of addition and multiplication (to do). We start out with a sentence of the form $\forall x \forall z \forall z' \in \mathbb{N}\ P(x, z, z', z'', z''')$ where P is quantifier free, equality is allowed, and we can reference arbitrary successors (i.e. $x+1$, $z+1$, $z'+1$,...) and prove the correctness for the IR of addition. Afterwards, we prove the correctness of the IR for multiplication, before eliminating certain quantifiers, replacing equality symbol with auxillary predicates, and replacing successors with predicates and/or treating them as Skolemized existential variables. The final result is a first order sentence of the form $\exists a \forall x \exists y \forall z \forall z'\ P(a, x, y, z, z')$.

2.1 Construction Idea

The basic idea is to construct a model using only binary primitive predicates, and of the form $\forall\exists\forall*$ (signature of the first order sentence). Natively, addition and multiplication equations lend themselves to ternary predicate representations:

$$\text{Add}(a, b, c) \iff a + b = c$$

$$\text{Mult}(a, b, d) \iff a \times b = d$$

However, we cannot treat these ternary predicates as primitives per our problem statement. The idea instead is to look at the arguments of a binary predicates not as operands, but as ways to store data.
In particular, we treat the columns of a binary predicate (second argument) as a data index to retrieve information about addition and multiplication operations.
Every addition rule $a + b = c$ should be stored in some column n. Similarly, every multiplication rule $a \times b = d$ should also be stored in some column m. For convenience and efficiency, we can use the same column n to store the operands of both addition and multiplication. We introduce binary predicates A, B, C, D for this.

$$\text{Add}(a, b, c) \iff (\exists n : A(a, n) \wedge B(b, n) \wedge C(c, n))$$

$$\text{Mult}(a, b, d) \iff (\exists n : A(a, n) \wedge B(b, n) \wedge D(d, n))$$

So far, this guarantees every addition and multiplication operation can be "stored" in some respective column. Ideally, we would simply want to "compute" addition and multiplication by addressing the operands (a, b) and getting the result (i.e. $c = a+b$, $d = a \times b$). The simplest idea to enforce this is to ensure that every column has at least some stored value of a, b, c, and d. Furthermore, to prevent conflict and simplify the proof of correctness, we will want to assert that such a, b, c, d are unqiue. We can write this as :

$$\forall n, \exists_1 a, b, c \; A(a, n) \wedge B(b, n) \wedge C(c, n) \wedge D(d, n)$$

($\exists_1$ is the uniqueness quantifier)

This way, every $n \in \mathbb{N}$ corresponds to exactly one addition and multiplication rule with the same operands a and b.

3 Arithmetic in FOL without equality

The idea of the previous section was to store an infinite "additive" and "multiplicative" "lookup table" for addition and multiplication. However, we will need to construct an explicit mapping that can map operands (a, b) to a column index n. Furthermore, first order statement of the type AEA* naturally only allow one successor reference (i.e. $x \to x + 1$) by performing Skolemization on the existential quantifier. From a geometric standpoint, we can interpret the model structure of AEA* as behaving like a 2-dimensional grid in the first quadrant of the coordinate plane. The columns represent horizontal movements, and the rows represent vertical movements.

With this in mind, we want to construct a function, or more specifically, a bijection (since $\mathbb{N}^{\nvDash}$ and $\mathbb{N}$ are both countable) where shifting (± 1) the operands moves to the next column (to the right) as much as possible. Let f define the mapping of a, b to its index column n.

3.1 Bijections

Let $f : \mathbb{N}^2 \to \mathbb{N}$ be a function from the set $\mathbb{N}^2$ to $\mathbb{N}$ defined by:

$$f(a,b) = \binom{a+b+1}{2} = \frac{(a+b)^2 + (a+b)}{2} + a = \frac{a^2 + 2ab + b^2 + a + b}{2} + a$$

The first few mappings defined by f are:

$$\begin{aligned}
f(0,0) &: 0 \\
f(0,1) &: 1 \\
f(1,0) &: 2 \\
f(0,2) &: 3 \\
f(1,1) &: 4 \\
f(2,0) &: 5 \\
f(0,3) &: 6 \\
f(1,2) &: 7 \\
f(2,1) &: 8 \\
f(3,0) &: 9 \\
f(0,4) &: 10
\end{aligned}$$

etc...

Intuitively, this appears to be a bijection from $\mathbb{N}^{\nvDash} \to \mathbb{N}$. In fact we will prove this because this is a crucial step in proving the correctness of an IR model of addition and multiplication that is constructed later.

Theorem 3.1. *f is a bijective function from $\mathbb{N}^2$ to $\mathbb{N}$.*

Proof. To show f is a bijection from $\mathbb{N}^2$ to $\mathbb{N}$. We must show that:

- f is onto (all natural numbers have the form $(a^2 + 2ab + b^2 + a + b)/2 + a$)
- f is one-to-one (all distinct pairs (a, b) map to different values)

We will show that f is onto.

By the well ordering principle, for every natural number n, there exists a smallest k such that
$$n < \binom{k+1}{2}$$
. Because k is the smallest integer such that $n < \binom{k+1}{2}$, it must be the case that $\binom{k}{2} \leq n$.

Because $\binom{k}{2} \leq n$, there is a non-negative integer r such that $\binom{k}{2} + r = n$.

Since $0 \leq n < \binom{k+1}{2}$, we have:

$$0 < \binom{k+1}{2}$$
$$0 < \frac{k^2+k}{2}$$
$$0 < k^2 + k$$
$$-k < k^2$$

If k is a natural number, then the equality holds for all $k > 0$. Hence, $k \geq 1$. We have from earlier:
$\binom{k}{2} + r = n$
So
$$\binom{k}{2} + r < \binom{k+1}{2}$$
$$r < \binom{k+1}{2} - \binom{k}{2}$$
$$r < k$$
(which follows Pascal's Identity)

Because a and b were defined to be arbitrary natural numbers, and $k \geq 1$, we can assert the existence of some a and b such that $a+b+1 = k$. Because r was an integer less than k (or $\leq k-1$), we have $r < a+b+1$, or $r \leq a+b$. Since a and b are arbitrary numbers that sum to $k-1$, and $r \leq k-1$, we can, without loss of generality, let $r = a$.
Hence, we have shown the existence of integers a and b such that for any n, $n = \binom{a+b+1}{2} + a$.

We now need to show that for any pairs of integers (a, b) and (c, d), $(f(a, b) = f(c, d)) \implies (a = c) \wedge (b = d)$ (i.e. injectivity).
Suppose that $f(a, b) = f(c, d) = n \in \mathbb{N}$.

We have $\binom{a+b+1}{2} + a = \binom{c+d+1}{2} + c$.

Let $u = a + b + 1$ and $v = c + d + 1$. We have:

$$\binom{u}{2} + a = \binom{v}{2} + c$$

Without loss of generality (by the well ordering principle), suppose that $u \leq v$. Let $u + w = v$ for some nonnegative integer w.

We have $a < u$ and $u \geq 1$. This, along with Pascal's Triangle Identity (which implies $\binom{u+1}{2} > \binom{u}{2}$ since $u > 0$), yields:

$$\binom{u}{2} + a \leq \binom{u}{2} + a + b \; (= \binom{u}{2} + u - 1) < \binom{u+1}{2} \leq \binom{u+i}{2}$$

(for any $i > 0$)

Since $u + w = v$ and $\binom{u}{2} + a = \binom{u+w}{2} + a$, but $\binom{u}{2} + a < \binom{u+i}{2} + a$ for $i > 0$, w must be 0, so $u = v$, and $a + b = c + d$.
Hence,

$$\binom{u}{2} + a = \binom{v}{2} + c = n \implies a = c$$

and

$$a + b = c + d \implies b = d$$

Since f is both onto and one-to-one or injective, f must be a bijection from $\mathbb{N}^2$ to $\mathbb{N}$. $\square$

4 IR of Addition in First Order Logic

To encode addition rules using binary predicates, we rely on the construction methods described in Section 2, in conjunction with the bijection f we outlined in Section 3. The idea is to use diagonal top-down (for the first operand a) and bottom-up (for the second operand b) encodings, to translate between different addition rules between successive columns (i.e. n to $n + 1$). When the bottom up rule (i.e. b) approaches zero (the first row), we then "slide" the sum c to the next value $c + 1$ and reset the top down counter a to zero, while setting b to $c + 1$, the the top-down and bottom-up loops repeat. Because a starts at 0 and traverses to c, and b starts at c and traverses to zero, and c consistently always shifts by one, every summation rule $a + b = c$ is uniquely addressable in a certain column. For instance, Column 0 might address the rule $0 + 0 = 0$ while column 4 might address the rule $1 + 1 = 2$.

4.1 The First Order Statement

Statement 4.1.

$$\forall x \forall z \forall z' \in \mathbb{N}:$$

$$\begin{aligned}
&A(0,0) \wedge \\
&B(0,0) \wedge \\
&C(0,0) \wedge \\
&((A(z,x) \wedge A(w,x)) \implies z = z') \wedge \\
&((B(z,x) \wedge B(w,x)) \implies z = z') \wedge \\
&((C(z,x) \wedge C(w,x)) \implies z = z') \wedge \\
&(C(x,z) \implies (C(x,z+1) \vee C(x+1,z+1))) \wedge \\
&((C(x,z) \wedge A(x,z)) \implies \neg A(x+1,z+1)) \wedge \\
&((\neg C(x,z) \wedge A(x,z)) \implies A(x+1,z+1)) \wedge \\
&(B(x+1,z) \implies B(x,z+1)) \wedge \\
&(B(0,x) \implies A(0,x+1)) \wedge \\
&((B(0,x) \wedge C(z,x)) \implies \neg C(z,x+1)) \wedge \\
&((\neg B(0,x) \wedge C(z,x)) \implies C(z,x+1)) \wedge \\
&((A(0,x) \wedge C(z,x)) \implies B(z,x)) \wedge
\end{aligned}$$

Theorem 4.1. *For all natural numbers n, there exists natural numbers a, b, c such that $a + b = c$ and $A(a,n) \wedge B(b,n) \wedge C(c,n)$. Furthermore, such a, b, c are unique.*

Proof. The theorem must be proven for all eligible pairs (n, a, b, c). To make the proof more efficient, we invoke the at-most-one rules in Statement 4.1, effectively showing the uniqueness of a, b, and c such that $A(a,n) \wedge B(b,n) \wedge C(c,n)$:

$$\begin{aligned}
&((A(z,n) \wedge A(z',n)) \implies z = z') \wedge \\
&((B(z,n) \wedge B(z',n)) \implies z = z') \wedge \\
&((C(z,n) \wedge C(z',n)) \implies z = z')
\end{aligned}$$

We can prove the existence criteria (that a, b, c actually exist), and the sum relationship ($a + b = c$) by induction. The remainder of the proof proceeds by induction on n. Suppose $S(n)$ is the statement we want to prove for all $n \in \mathbb{N}$:

$$S(n) =$$

$$\exists a, b, c : A(a,n) \wedge B(b,n) \wedge C(c,n)$$

and

$$(A(a,n) \wedge B(b,n) \wedge C(c,n)) \implies (a+b=c)$$

The initial case $S(0)$ is trivially met by the axioms of Statement 4.1 (where $n,a,b,c=0$):

$$A(0,0) \wedge B(0,0) \wedge C(0,0)$$

and

$$a+b=c \text{ is true : } 0+0=0$$

For the induction on n, Suppose that $S(n)$ is true. Let (a,b,c) be the unique pair satisfying $S(n)$ [inductive hypothesis]. Let (a',b',c') be the unique pair satisfying $S(n+1)$, if it exists. We divide the proof into two cases based on whether or not $B(0,n)$ is true or false.

First assume $B(0,n)$ is false. Because we have the existence of a b such that $B(b,n)$ it must be the case that $b>0$, and so the rule (letting $x+1=b$):

$$B(b,z) \implies B(b-1,z+1)$$

applies (as $b-1$ is nonnegative). So $B(b-1,n+1)$ is true.
Additionally (from the falsity of $B(0,x)$), the rule

$$((\neg B(0,n) \wedge C(c,n)) \implies C(c,n+1))$$

also applies (letting $x=n$ and $z=c$). Hence, $C(c,n+1)$ is true.

Now letting $x=a$ and $z=n$, we obtain the following two implications:

$$((C(a,n) \wedge A(a,n)) \implies \neg A(a+1,n+1))$$

$$((\neg C(a,n) \wedge A(a,n)) \implies A(a+1,n+1))$$

If $C(a,n)$ were true, then $c=a$ by uniqueness constraints, and $a+b=c=a$, which requires $b=0$, which contradicts that $b>0$. Therefore, $C(a,n)$ must be false (and $a \neq c$), in which case, the second implication rule applies, while the first is vacuously true. By the second rule, we have $A(a+1,n+1)$ to be true.
By letting $a'=a+1$, $b'=b-1$, $c'=c$ have found a pair (a',b',c') such that:

$$A(a',n+1) \wedge B(b',n+1) \wedge C(c',n+1)$$

and

$$a' + b' = c' \text{ is true : } a' + b' = a + 1 + b - 1 = a + b = c = c'$$

From the uniqueness of a', b', c', no such other pairs exist and the statement S holds for $n + 1$ when $B(0, n)$ is false.

Now assume, for the second case, that $B(0, n)$ is true. In that case, it follows directly from the rule

$$B(0, n) \implies A(0, n + 1)$$

in Statement 4.1 that $A(0, n + 1)$ is also true. By the rule (substituting $x = n$ and $z = c$ in the original statement):

$$(B(0, n) \wedge C(c, n)) \implies \neg C(c, n + 1)$$

and by the rule

$$C(c, n) \implies (C(c, n + 1) \vee C(c + 1, n + 1))$$

we can conclude that since $C(c, n + 1)$ is false (first rule listed above), $C(c + 1, n + 1)$ must be true (second rule listed above). Using the rule ($z = c + 1$, $x = n + 1$):

$$(A(0, n + 1) \wedge C(c + 1, n + 1)) \implies B(c + 1, n + 1)$$

in conjunction with the previously stated two, we can conclude that $B(c + 1, n+1)$ is true. Hence we have the existence of elements $a' = 0$, $b' = c+1$, and $c' = c + 1$ such that:

$$A(a', n + 1) \wedge B(b', n + 1) \wedge C(c', n + 1)$$

and

$$a' + b' = c' \text{ is true : } 0 + (c + 1) = c + 1$$

By the at most one constraint of a', b', c', $S(n+1)$ must therefore be true (since there are no other (a', b', c') pairs to prove).
Therefore, since the statement $S(n + 1)$ is true in both cases (depending on whether $B(0, n)$ is true or false), it holds. □

Theorem 4.2. *For all natural numbers n, a, b, $A(a, n)$ and $B(b, n)$ are true if and only if $n = f(a, b)$.*

Proof. In the previous theorem, we established that there exists a unique a and b such $A(a, n)$ and $B(b, n)$ are true, for every n. By Theorem 3.1, because f is a bijection, there must exist unique u and v such $n = f(u, v)$. The goal is to show that $u = a$ and $v = b$.
We can prove this by induction:

$$S(n) = \{\forall a, b, u, v \in \mathbb{N}\,((f(u, v) = n) \wedge A(a, n) \wedge B(b, n)) \implies ((u = a) \wedge (v = b))\}$$

For $S(0)$, we previously established that $(a,b,c) = (0,0,0)$ is the unique pair (a,b,c).
On the other hand if $f(u,v) = \binom{u+v+1}{2} + u = \frac{u^2+2uv+v^2+u+v}{2} + u$ is zero, then $u = 0$ and $v^2 + v = 0 \implies v = 0$. Hence $u = a = 0$ and $v = b = 0$.
For the inductive step, we can, similar to the previous theorem, split the step into two cases.
The inductive hypothesis assumes that $u = b$ and $v = a$.
As with the previous proof let (a,b,u,v) be the pair that satisfies $S(n)$. Let $c = a+b$. Let (a',b') be the pair satisfying $A(a', n+1) \wedge B(b', n+1)$, $c' = a' + b'$, and (u',v') be the pair satisfying $f(u',v') = n+1$. Both (a',b') and (u',v') must exist by Theorems 4.1 and 3.1, respectively. The objective is to show they are equal.

Case I: $b \neq 0$.
In this case, since b is a natural number, $b > 0$. It was shown in Theorem 4.1 that for $n+1$, both $A(a+1, n+1)$ and $B(b-1, n+1)$ are both true. Hence $a' = a+1$ and $b' = b+1$. Because of the uniqueness of solutions to $f(u',v') = n+1$, it is sufficient to show that $f(a+1, b-1) = n$. (which would imply that $u = a' = a+1$ and $v = b' = b-1$.
Using the binomial form:

$$f(a+1, b-1) = \binom{a+1+b-1+1}{2} + a + 1 = \binom{a+b+1}{2} + a + 1 =$$

$$(\binom{a+b+1}{2} + a) + 1 =$$

$$n+1$$

Hence, since $f(a+1, b-1) = n+1$, $u = a' = a+1$ and $v = b' = b-1$ and the induction step is proven.

Case II : $b = 0$.
In this case, we have $f(a,0) = \binom{a+1}{2} + a = \frac{a^2+a}{2} + a = \frac{a^2+3a}{2} = n$. From the inductive reasoning of Theorem 4.1, it was shown that for $n+1$, $A(0, n+1)$ and $B(c+1, n+1)$, and consequently $C(c+1, n+1)$ is true $(a' = 0, b' = c+1, c' = c+1)$. As $f(u',v') = n+1$, it is sufficient to show that $f(0, c+1) = f(0, a+b+1)$ evaluates to $n+1$:

$$f(0, c+1) = \binom{0 + (c+1) + 1}{2} + 0 = \binom{c+2}{2}$$

Yet by the induction Hypothesis, $f(a,b) = \binom{a+b+1}{2} + a = \binom{c+1}{2} + a = n$.

Using Pascal's Identity, we have:

$$\binom{c+2}{2} - \binom{c+1}{2} = c+1 = a+b+1$$

So we can express

$$\binom{c+2}{2} =$$

$$\binom{c+1}{2} + c + 1 =$$

$$\binom{a+b+1}{2} + a + b + 1 =$$

$$\binom{a+1}{2} + a + 1 =$$

$$\text{(because } b = 0\text{)}$$

$$(\binom{a+1}{2} + a) + 1 =$$

$$f(a,b) + 1 =$$

$$n + 1$$

$$\text{(applying the inductive hypothesis)}$$

Hence, $f(0, c+1) = \binom{c+2}{2} = n+1$, so $u' = a' = 0$ and $v' = b' = c+1$.
In either case ($b \neq 0$) or $b = 0$, the inductive step holds and $S(n+1)$ is true.

$\square$

Corollary 4.1. *For all natural numbers a, b, there exist exactly one natural number n such that $A(a,n)$, $B(b,n)$, and consequently, $C(a+b,n)$ are all true.*

This directly follows from the reasoning Theorem 4.1 to show they exist, the reasoning of Theorem 4.2 to establish the relationship of a, b to f, and Theorems 3.1 and 4.1 to show that such a and b are unique, and again, Theorem 4.1 to show that if $C(c,n)$ is true, $c = a+b$.

5 IR of Multiplication in First Order Logic

In the addition representation, we were able to diagonally "shift" the operands a and b in opposite directions to enumerate all integers a, b that sum to any given integer c, and when the enumeration was done (i.e. $b = 0$), we increment c and repeat the process forever.
Unfortunately, multiplicative properties are difficult to enumerate in a sequential column fashion. Instead, now that we have additive properties established for A, B, and C, we can use those to encode multiplication only using recursive

addition rules. The idea is to use the fact that $a \times b$ is equivalent to adding a, b number of times. The recursive solution is as follows:

$$\begin{cases} a \times b = 0, & \text{if } b = 0 \\ a \times b = a \times (b-1) + a, & \text{if } b > 0 \end{cases}$$

To implement this behavior in first order logic, we objectively want to encode the property

$$(A(a,n) \wedge B(b,n) \wedge D(d,n)) \implies a \times b = d$$

By Theorem 4.2, we already have the operands a and b encoded into a unique column n (and their sum c), so the only remaining objective is to encode their product d, into the same column. As with A, B, C, we will want to establish uniqueness for D. We will also need an intermediate variable P to represent the previous product (i.e. $p = a \times (b-1)$) and then address the column which computes the sum of a and p, and then assert that the product of a and b is equal to the sum of a and p. We describe the first order statement below. We use z_2, z_3, z_4 as extra quantifiers instead of the extra z' quantifier (variable naming convention change). The conjunctions contained in the first set of square brackets are the addition rules we proved hold in Section 4. The conjunctions contained in the second set of square brackets are the multiplication rules we intend to encode and prove hold as intended.

5.1 The First Order Statement

Statement 5.1.

$$\forall x \forall z \forall z_2 \forall z_3 \forall z_4 \in \mathbb{N} :$$

$$
\begin{aligned}
&[A(0,0) \wedge \\
&B(0,0) \wedge \\
&C(0,0) \wedge \\
&((A(z,x) \wedge A(z_2,x)) \implies z = z_2) \wedge \\
&((B(z,x) \wedge B(z_2,x)) \implies z = z_2) \wedge \\
&((C(z,x) \wedge C(z_2,x)) \implies z = z_2) \wedge \\
&(C(x,z) \implies (C(x,z+1) \vee C(x+1,z+1))) \wedge \\
&((C(x,z) \wedge A(x,z)) \implies \neg A(x+1,z+1)) \wedge \\
&((\neg C(x,z) \wedge A(x,z)) \implies A(x+1,z+1)) \wedge \\
&(B(x+1,z) \implies B(x,z+1)) \wedge \\
&(B(0,x) \implies A(0,x+1)) \wedge \\
&((B(0,x) \wedge C(z,x)) \implies \neg C(z,x+1)) \wedge \\
&((\neg B(0,x) \wedge C(z,x)) \implies C(z,x+1)) \wedge \\
&((A(0,x) \wedge C(z,x)) \implies B(z,x))] \\
&\wedge \\
&[((D(z,x) \wedge D(z_2,x)) \implies z = z_2) \wedge \\
&((P(z,x) \wedge P(z_2,x)) \implies z = z_2) \wedge \\
&(B(0,x) \implies D(0,x)) \wedge \\
&(B(0,x) \implies \neg P(z,x)) \wedge \\
&((A(z,z_3) \wedge B(x,z_3) \wedge D(z_2,z_3) \wedge A(z,z_4) \wedge B(x+1,z_4)) \implies P(z_2,z_4)) \wedge \\
&((A(x,z_2) \wedge P(z,z_2) \wedge A(x,z_3) \wedge B(z,z_3) \wedge C(z_4,z_3)) \implies D(z_4,z_2))]
\end{aligned}
$$

Theorem 5.1. *For all natural numbers n, there exists natural numbers a, b, d such that $a \times b = d$ and $A(a,n) \wedge B(b,n) \wedge D(d,n)$. Furthermore, such a, b, d are unique.*

Proof. From Theorem 4.1, we already know there exists a and b such that $A(a,n) \wedge B(b,n)$ are both true. Therefore, the only remaining objective is to show that $D(d,n)$ is true iff $a \times b = d$.
It helps to establish the uniqueness rules for D and P ahead of time:

$$((D(z,x) \wedge D(z_2,x)) \implies z = z_2)$$

$$((P(z,x) \wedge P(z_2,x)) \implies z = z_2)$$

Hence, we now only have to show that for this specific d, it exists, and is the product of a and b.
Rather than inducting in the columns as we did for the proof of theorem 4.1, we will instead induct on the value of b, treating a as an arbitrary fixed integer.
Let S be the following statement:

$$S(b) : (\forall a \forall d\ (A(a,n) \wedge B(b,n) \wedge D(d,n)) \implies (a \times b = d))$$

S(0) reduces down to $(\forall a \forall d\ (A(a,n) \wedge B(0,n) \wedge D(d,n)) \implies (a \times 0 = d))$ so for the statement to hold, $d = 0$. We have the implication rule

$$(B(0,n) \implies D(0,n))$$

(substituting $x = n$) so for the implication to follow, $d = 0$ (which is true if $b = 0$) and the consequent $a \times 0 = 0$ is true, so $S(0)$ is true. If $d \neq 0$, then $D(d,n)$ is false and $S(0)$ vacuously true. In either case, $S(0)$ is true.
Now assume $S(b)$ is true. For a fixed $b > 0$ (we just showed $b = 0$ yields a true statement), we have the rule

$$((A(z,z_3) \wedge B(x,z_3) \wedge D(z_2,z_3) \wedge A(z,z_4) \wedge B(x+1,z_4)) \implies P(z_2,z_4))$$

Making the substitutions $z = a$, $x = b$, $z_2 = d$, $z_3 = n$, $z_4 = m$, this becomes:

$$((A(a,n) \wedge B(b,n) \wedge D(d,n) \wedge A(a,m) \wedge B(b+1,m)) \implies P(d,m))$$

By the induction hypothesis, $a \times b = d$. The unique column m exists operands a and $b+1$, and we directly assert $P(a \times b, m)$ to be true. Since true values of P (for each column) are unique, we know that P uniquely encodes the previous product into the row addressing the successor of the second operand $b+1$. By the rule

$$((A(x,z_2) \wedge P(z,z_2) \wedge A(x,z_3) \wedge B(z,z_3) \wedge C(z_4,z_3)) \implies D(z_4,z_2))$$

Making the substitutions $x = a$, $z = d$, $z_2 = m$, $z_3 = m'$, $z_4 = a + d$, this becomes:

$$((A(a,m) \wedge P(d,m) \wedge A(a,m') \wedge B(d,m') \wedge C(a+d,m')) \implies D(a+d,m))$$

From the previous rule and deduction steps, $P(d,n)$ was implied to be true, $A(a,m)$ was true by assumption, there is a unique column m' containing operands a and d, respectively, and their sum $a+d$ (predicate C). Hence $D(a+d,m)$ is true.
Because m was assumed to be the unique column variable holding a and $b+1$ as operands (i.e. $A(a,m)$ and $B(b+1,m)$), and $a + d = a + a \times b = a \times (b+1)$, it follows that the statement $S(b+1)$ is true (since the value of d' such that $D(d',m)$ is true, is the product of a and $b+1$).

□

In later sections [TODO], we will describe how to rewrite statement 5.1 using less quantifiers (quantifier elimination), at the cost of only adding a few more intermediate predicates, eliminating equality comparisons (uniqueness constraints), eliminating successor references to $z+1$, and eliminating the use of constant 0 (other than $(0,0)$).

6 Equality, Successor, and Constant Eliminations

In this section, we turn our attention to the uniqueness constraints. When we mention our First-Order statement of interest, we will be referring to Statement 5.1.
The uniqueness constraints we imposed are:

$$((A(z,x) \wedge A(z_2,x)) \implies z = z_2)$$
$$((B(z,x) \wedge B(z_2,x)) \implies z = z_2)$$
$$((C(z,x) \wedge C(z_2,x)) \implies z = z_2)$$
$$((D(z,x) \wedge D(z_2,x)) \implies z = z_2)$$
$$((P(z,x) \wedge P(z_2,x)) \implies z = z_2)$$

The goal is to rewrite each of these statements in predicate form, (i.e. replacing $z = z_2$ with a predicate $E(z,z_2)$). However, we cannot simply assert the definition of an equality predicate E from $=$. To do this, we encode both equality and successor references using predicates E and S respectively, which allows as us to not only solve the problem of removing equality symbols, but also the problem of removing extra successor references like $z+1$ (which aren't allowed in the prefix $\forall\exists\forall*$ form). The references to $z+1$ we want to remove are:

$$C(x,z) \implies (C(x,z+1) \vee C(x+1,z+1))$$
$$(C(x,z) \wedge A(x,z)) \implies \neg A(x+1,z+1)$$
$$(\neg C(x,z) \wedge A(x,z)) \implies A(x+1,z+1)$$
$$B(x+1,z) \implies B(x,z+1)$$

At first glance, it is not obvious how we can replace $z+1$ only using z and additional predicates. We explain the idea below. In more generality, suppose we have a statement of the form:
$Q(z+1)$
for some $z \in \mathbb{N}$
where Q is an arbitrary predicate (we avoid using the variable P).
If we introduce an auxillary predicate S such that $S(z,z_2)$ is true if and only if $z_2 = z+1$, we can write Q (preserving equivalence) as:

$$S(z,z_2) \implies Q(z_2)$$

The consequent (right side) of the implication is only forced to be true if $z_2 = z + 1$. If $z_2 \neq z + 1$, then the antecedent (left side) is false, so the implication is vacously true. This does not mean $Q(z_2)$ has to be true, it just means that the value of $Q(z_2)$ has no influence on the truth of the new implication. Hence logical equivalence is preserved.
Eliminating the constant term 0 (other than $(0, 0)$) requires referencing a point of origin (or row/column of origin). We want to eliminate other explicit non-origin references to zero because the $\forall\exists\forall*$ structure does not permit for them. In theory, we would actually use $\exists\forall\exists\forall*$, however, the first $\exists$ quantifier is just to assert the domain is non-empty in formal logic, it shouldn't be taken as a global reference to 0. Ideally, we intend to eliminate literals like $A(0, x)$ or $B(0, x)$.

Using these three reductions, we introduce predicates E and S which allow us to address successor and equality operations without explicitly using them in our sentence. We describe this idea below.

6.1 Equality and Successor statement

Before describing the exact statement, it is useful, from a conceptual standpoint, how we can encode explicit equality and successor relationships without ever using $=$ or $z + 1$. Let $E(x, z)$ be the predicate representing equality (i.e. $x = z$) and $S(x, z)$ be the predicate representing successor references (i.e. $z = x + 1$). We initially assert that $E(0, 0)$ as $0 = 0$. We then want the coordinate directly to the right of E i.e. $(x, x+1)$ to be true for S. By moving downward, we get the coordinate $(x + 1, x + 1)$, and E should be true for this value. By repeating this process infinitely often, we enforce E and S are true for the coordinates that we want them to be true for. To enforce uniqueness and eliminate ambiguity, we use an auxiliary predicate S' to occupy the remaining space of the upper octant strictly above the main diagonal i.e. $S'(x, z), x < z - 1$ that S does not cover, and lastly, enforce mutual exclusion (i.e. disjoint) to only allow at most one of $E(x, z)$, $S(x, z)$, $S'(x, z)$ to be true at a time.

Statement 6.1.

$$\forall x \forall z \forall z' \in \mathbb{N} :$$

$$
\begin{aligned}
E(0,0)\ \wedge \\
\neg(S(x,z) \wedge E(x,z))\ \wedge \\
\neg(S(x,z) \wedge S'(x,z))\ \wedge \\
\neg(E(x,z) \wedge S'(x,z))\ \wedge \\
(S(x+1,z) \implies S'(x,z))\ \wedge \\
(S'(x+1,z) \implies S'(x,z))\ \wedge \\
(E(z,x) \implies S(z,x+1))\ \wedge \\
(S(x,z) \iff E(x+1,z))\ \wedge
\end{aligned}
$$

Theorem 6.1. *For all natural numbers n, m, the following hold:*

$$E(n,m) \iff (m = n)$$

$$S(n,m) \iff (m = n+1)$$

Proof. We proceed by induction on m.
Let

$$T(m) = \text{for all } m \in \mathbb{N} : (E(n,m) \iff (m = n)) \wedge (S(n,m) \iff (m = n+1))$$

For $m = 0$, For $T(0)$ to be true, $E(0,0)$ must be true, which it is by axiomatic assumption. For other non-zero n, $(E(n,m) \iff (m = n))$ means $E(n,0)$ must be false. Additionally, because $m = 0 \neq n+1$ for any natural number n, $S(n,0)$ must be false for all n.
Suppose then for sake of contradiction, that there exists an n such that $S(n,0)$ is true. Such an n must be greater than zero, because of the rule

$$\neg(S(x,z) \wedge E(x,z))$$

and the fact that $E(0,0)$ is true. Hence the rule

$$(S(x+1,z) \implies S'(x,z))$$

applies, and setting $x+1 = n$, $S'(n-1,z)$ is false. Because of the disjoint rules:

$$\neg(S(x,z) \wedge S'(x,z))$$
$$\neg(E(x,z) \wedge S'(x,z))$$

it must be the case that $E(n-1,z)$ and $S(n-1,z)$ are both false. If $n = 1$ and $z = 0$, then $S'(0,0)$ is true, so $E(0,0)$ is false, which contradicts that it is true.

Therefore $n > 1$, and the rule

$$(S'(x+1,z) \implies S'(x,z))$$

applies. By letting $x+1 = n-1$ and $z = 0$, we get $S'(n-2,0)$ is false. More generally, if $S'(n-k,0)$ is true and $k > n$, then the rule applied to $x+1 = n-k$ and $z = 0$ leads to the conclusion that for all $k < n$, $S'(n-k,0)$ is true.
Letting $k = 1$ and $z = 0$, the implication rule leads to the conclusion that $S'(1,0) \implies S'(0,0)$. This is a contradiction, (which would force $E'(0,0)$ to be false, since $S'(x,z)$ and $E'(x,z)$ cannot be true simultaneously).
Hence, $S(n,0)$ is false for all n.
Now suppose there exists a non-zero $n > 0$ such that $E(n,0)$ is true. Since $n = x+1$ for some natural number x, the bi-conditional implication rule

$$(S(x,z) \iff E(x+1,z))$$

is applicable, forcing $S(n-1,z)$ to be true. However, this contradicts the fact that $S(n,z)$ is false for all natural numbers n.
Therefore,

$$(E(n,m) \iff (m = n))$$

holds because $E(n,0)$ is only true for $n = 0$, and both sides are false otherwise.

$$(S(n,m) \iff (m = n+1))$$

holds because for the right hand side, $m = 0$ and $0 = n+1$ is an impossibility, meanwhile the left hand side was shown to be false for all n.
Therefore $T(0)$ is true.

We can now start the inductive step. Assume that, $T(m)$ is true.
By the induction hypothesis, we have

$$S(m-1,m) \text{ if } (m > 0)$$

$$E(m,m)$$

both be true, but no other other values in the same column n will make the predicates true. By the implication rule:

$$(E(z,x) \implies S(z,x+1))$$

$S(m,m+1)$ must be true. Then by the bidirectional implication rule

$$(S(x,z) \iff E(x+1,z))$$

$E(m+1,m+1)$ must be true. To show uniqueness (the "only if" part of T), suppose that $S(n', m+1)$ was true for some $n' \neq m$. In that case, we can reuse the same reasoning as when $m = 0$:

Because $S(m, m+1)$ is true, $S'(m-1, m+1)$ (if $m > 0$, if $m = 0$, then this step doesn't apply) and consequently $S'(k, m+1)$ must be true for any $k < m$, forcing $n' > m$. However, using yet the same reasoning, if $S(n', m+1)$ is true, then so is $S'(k, m+1)$ for any $k < n'$. Since $m < n'$, $S'(m, m+1)$ is true, which means $S(m, m+1)$ cannot be true, yielding a contradiction. Therefore, $n' = m$ and the uniqueness of true values of S in column $m+1$ is preserved.

The uniqueness of values in column $m+1$ such that E is true follows from the uniqueness of S. Suppose there exists a value n' such that $n' \neq m+1$ and $E(n', m+1)$ is true. From the bidirectional implication rule outlined earlier, $S(n'-1, m+1)$ is true (if $n' > 0$), and because $n' \neq m+1$, $n'-1 \neq m$, so $S(n'', m+1)$ is true for some value other than $n'' = m$, which is a contradiction. Because the uniqueness was preserved, the induction holds and $T(m)$ is true.

□

6.2 Row of Origin statement

We introduce a new predicate Z to reference the entire first row (i.e. row at index 0). The idea is to encode a predicate which is true for $Z(x, z)$ iff $x = 0$, and false otherwise.

Statement 6.2.

$$\forall x \forall z \in \mathbb{N} :$$

$$\begin{aligned} Z(0,0) \ \wedge \\ \neg Z(x+1, z) \ \wedge \\ (Z(z,x) \implies Z(z, x+1)) \end{aligned}$$

Theorem 6.2. $\forall n, m \in \mathbb{N} : Z(n,m) \iff (n = 0)$

Proof. By the row implication

$$(Z(z,x) \implies Z(z, x+1))$$

once a value of x exists such that $Z(z,x)$ is true, then $Z(z,w)$ is true for all $w \geq x$ from an induction standpoint. Because $Z(0,0)$ is true, so is $Z(0,1), Z(0,2), Z(0,3), \ldots$. If $n > 0$, then $n = x+1$ for some natural number x, and the literal

$$\neg Z(x+1, z)$$

directly prohibits any value of $Z(n,m)$ from ever being true. □

If we have something like $\forall x B(0, x)$ as a statement, and the goal is to eliminate the 0 constant term in B using Z, we can just reference 0 with predicate Z with an additional quantifier like $\forall x \forall z\ (Z(z, x) \implies B(z, x))$. Because $Z(z, x)$ is only ever true if $z = 0$, the implication consequence only follows conditionally upon $z = 0$, and is vacuously true otherwise (meaning the truth value of B is irrelevant and preserved).

6.3 Re-expressing Equality, Successors, and Zero

With the information proven regarding E, S, and Z, and using the conjunction of Statements 5.1 and 6.1, we can rewrite the entire first order statement as.

Statement 6.3.

$$\forall x \forall z \forall z_2 \forall z_3 \forall z_4 \in \mathbb{N} :$$

$$
\begin{aligned}
&[A(0,0) \wedge \\
&B(0,0) \wedge \\
&C(0,0) \wedge \\
&((A(z,x) \wedge A(z_2,x)) \implies E(z,z_2)) \wedge \\
&((B(z,x) \wedge B(z_2,x)) \implies E(z,z_2)) \wedge \\
&((C(z,x) \wedge C(z_2,x)) \implies E(z,z_2)) \wedge \\
&((C(x,z) \wedge S(z,z_2)) \implies (C(x,z_2) \vee C(x+1,z_2))) \wedge \\
&((C(x,z) \wedge A(x,z) \wedge S(z,z_2)) \implies \neg A(x+1,z_2)) \wedge \\
&((\neg C(x,z) \wedge A(x,z) \wedge S(z,z_2)) \implies A(x+1,z_2)) \wedge \\
&((B(x+1,z) \wedge S(z,z_2)) \implies B(x,z_2)) \wedge \\
&((Z(z,x) \wedge B(z,x)) \implies A(z,x+1)) \wedge \\
&((Z(z_2,x) \wedge B(z_2,x) \wedge C(z,x)) \implies \neg C(z,x+1)) \wedge \\
&((Z(z_2,x) \wedge \neg B(z_2,x) \wedge C(z,x)) \implies C(z,x+1)) \wedge \\
&((Z(z_2,x) \wedge A(z_2,x) \wedge C(z,x)) \implies B(z,x))] \\
&\wedge \\
&[((D(z,x) \wedge D(z_2,x)) \implies E(z,z_2)) \wedge \\
&((P(z,x) \wedge P(z_2,x)) \implies E(z,z_2)) \wedge \\
&((Z(z,x) \wedge B(z,x)) \implies D(z,x)) \wedge \\
&((Z(z_2,x) \wedge B(z_2,x)) \implies \neg P(z,x)) \wedge \\
&((A(z,z_3) \wedge B(x,z_3) \wedge D(z_2,z_3) \wedge A(z,z_4) \wedge B(x+1,z_4)) \implies P(z_2,z_4)) \wedge \\
&((A(x,z_2) \wedge P(z,z_2) \wedge A(x,z_3) \wedge B(z,z_3) \wedge C(z_4,z_3)) \implies D(z_4,z_2))] \\
&\wedge \\
&[E(0,0) \wedge \\
&\neg(S(x,z) \wedge E(x,z)) \wedge \\
&\neg(S(x,z) \wedge S'(x,z)) \wedge \\
&\neg(E(x,z) \wedge S'(x,z)) \wedge \\
&(S(x+1,z) \implies S'(x,z)) \wedge \\
&(S'(x+1,z) \implies S'(x,z)) \wedge \\
&(E(z,x) \implies S(z,x+1)) \wedge \\
&(S(x,z) \iff E(x+1,z))] \\
&\wedge \\
&[Z(0,0) \wedge \\
&\neg Z(x+1,z) \wedge \\
&(Z(z,x) \implies Z(z,x+1))]
\end{aligned}
$$

7 Quantifier Eliminations

In this section, we describe how to eliminate the z_3 and z_4 quantifiers by rewriting the logical statements

$$((A(z,z_3) \wedge B(x,z_3) \wedge D(z_2,z_3) \wedge A(z,z_4) \wedge B(x+1,z_4)) \implies P(z_2,z_4))$$

and

$$((A(x,z_2) \wedge P(z,z_2) \wedge A(x,z_3) \wedge B(z,z_3) \wedge C(z_4,z_3)) \implies D(z_4,z_2))$$

In particular, we will focus on the idea behind eliminating the first quantifier z_4 before turning attention to the second one, z_3. Again, by "eliminating" quantifiers, we mean re-writing the sentence in a way that uses one less quantifier bound variables.

7.1 Eliminating the First Quantifier

For this subsection, we will be referring to rules/axioms in Statement 6.2. To understand how to eliminate the first quantifier, it is good to understand the logic behind the recursive multiplicative statements. Regarding the first rule, all we are doing is retrieving the column n that contains the product rule $a \times b = d$, and storing that result in the column m containing a and $b+1$, as the previous product p (i.e. from the perspective of m, $p = a \times b$).

In the second rule, we look at an arbitrary column n (fresh reset here since we are talking about a global rule), and if there is a value p stored in that column (from the perspective of n), it has the value of $a \times (b-1)$, and we then retrieve the column m, which contains the operands a and $(b =)p$ (so for column m, the operands are a and p), and retrieve the sum $(c =)a + p$. We then store this result in column n as the designated d value $d = a + p$, which is equivalent to $d = a + a \times (b-1) = a \times b$.

In summary, if we keep track locally of variables:
Let $A(a,n), B(b,n), D(d,n)$ be true.
So $a \times b = d$ is true.
Let m be the column containing $A(a,m)$ and $B(b+1,m)$ to be true.
We then store the value of d, $(D(d,n))$ as the previous result in column m, as $P(d,m)$.
Let n' be the column containing $A(a,n')$ and $B(d,n')$ as true values. Then we retrieve the sum, $a+d$ $(C(a+d,n'))$, and finally, we store this result back in column m as the product of $a+d = a \times (b+1)$, so $D(a+d,m)$ is true.

The idea with eliminating quantifiers is to preserve this same logic, keeping A, B, C, D, P logically intact, while using extra auxiliary steps to retrieve the "locations" of intermediate operations. Looking at the antecedent of the first statement

$$((A(z, z_3) \wedge B(x, z_3) \wedge D(z_2, z_3) \wedge A(z, z_4) \wedge B(x+1, z_4)) \implies P(z_2, z_4))$$

we do not have enough quantifiers to retrieve A, B, D and the column containing $b+1$ simultaneously. We can reduce a quantifier in this step at the cost of requiring two steps instead of one to store the result of the previous product (relative to the successor column).
We do this by addressing the location of the column m containing operands a and $b+1$, then storing that column m as information in the original column n. We then address the previous product (relative to m), and use the information of m (which is "stored" at location n), to assign the value of p relative to m as the product d. This logic can be described with the following set of sentences. Here, we will use temporary markers for quantifiers (v, v_1, v_2) to avoid confusion with the original four pointers (z, z_1, z_2, z_3):

$$\begin{aligned} ((R(v, x) \wedge R(v_2, x)) \implies E(v, v_2)) \wedge \\ ((A(v, v_2) \wedge B(x, v_2) \wedge A(v, v_3) \wedge B(x+1, v_3)) \implies R(v_3, v_2)) \wedge \\ ((D(x, v) \wedge R(v_2, v)) \implies P(x, v_2)) \end{aligned}$$

Here, we introduced a new predicate R to store the information about the column containing a and $b+1$ in the column containing a and b, We then used that information to store the product in the successor (i.e. relative to b only) column as the previous product.
We apply this two-step reasoning once again to retrieve the column containing a and $(b =)p$ (storing it as data in the column containing a and $b+1$ as operands), then store the sum (i.e. $a+p$) as the product of a and $b+1$ by referencing the original column containing a and $b+1$ from the perspective of the column containing a and p as operands.

$$\begin{aligned} ((Z(z_2, x) \wedge B(z_2, x)) \implies \neg U(z, x)) \wedge \\ ((U(v, x) \wedge U(v_2, x)) \implies E(v, v_2)) \wedge \\ ((A(v, v_2) \wedge P(x, v_2) \wedge A(v, v_3) \wedge B(x, v_3)) \implies U(v_3, v_2)) \wedge \\ ((C(x, v) \wedge U(v, v_2)) \implies D(x, v_2)) \end{aligned}$$

We enforced the exclusion of true values of U occurring in a column where is $b = 0$, as with P, there is no previous product stored in such a column (as it is a base case), so there is also no need to store the intermediate U either, so it is necessary to prevent unwanted ambiguity.
With this information in mind, we can replace $v = z$, $v_2 = z_2$, and $v_3 = z_3$ and

re-write the multiplication clauses as follows:

Statement 7.1.

$$\forall x \forall z \forall z_2 \forall z_3 \in \mathbb{N} :$$

$$
\begin{aligned}
&[A(0,0) \wedge \\
&B(0,0) \wedge \\
&C(0,0) \wedge \\
&((A(z,x) \wedge A(z_2,x)) \implies E(z,z_2)) \wedge \\
&((B(z,x) \wedge B(z_2,x)) \implies E(z,z_2)) \wedge \\
&((C(z,x) \wedge C(z_2,x)) \implies E(z,z_2)) \wedge \\
&((C(x,z) \wedge S(z,z_2)) \implies (C(x,z_2) \vee C(x+1,z_2))) \wedge \\
&((C(x,z) \wedge A(x,z) \wedge S(z,z_2)) \implies \neg A(x+1,z_2)) \wedge \\
&((\neg C(x,z) \wedge A(x,z) \wedge S(z,z_2)) \implies A(x+1,z_2)) \wedge \\
&((B(x+1,z) \wedge S(z,z_2)) \implies B(x,z_2)) \wedge \\
&((Z(z,x) \wedge B(z,x)) \implies A(z,x+1)) \wedge \\
&((Z(z_2,x) \wedge B(z_2,x) \wedge C(z,x)) \implies \neg C(z,x+1)) \wedge \\
&((Z(z_2,x) \wedge \neg B(z_2,x) \wedge C(z,x)) \implies C(z,x+1)) \wedge \\
&((Z(z_2,x) \wedge A(z_2,x) \wedge C(z,x)) \implies B(z,x))] \\
&\wedge \\
&[((D(z,x) \wedge D(z_2,x)) \implies E(z,z_2)) \wedge \\
&((P(z,x) \wedge P(z_2,x)) \implies E(z,z_2)) \wedge \\
&((R(z,x) \wedge R(z_2,x)) \implies E(z,z_2)) \wedge \\
&((U(z,x) \wedge U(z_2,x)) \implies E(z,z_2)) \wedge \\
&((Z(z,x) \wedge B(z,x)) \implies D(z,x)) \wedge \\
&((Z(z_2,x) \wedge B(z_2,x)) \implies \neg P(z,x)) \wedge \\
&((Z(z_2,x) \wedge B(z_2,x)) \implies \neg U(z,x)) \wedge \\
&((A(z,z_2) \wedge B(x,z_2) \wedge A(z,z_3) \wedge B(x+1,z_3)) \implies R(z_3,z_2)) \wedge \\
&((D(x,z) \wedge R(z_2,z)) \implies P(x,z_2)) \wedge \\
&((A(z,z_2) \wedge P(x,z_2) \wedge A(z,z_3) \wedge B(x,z_3)) \implies U(z_3,z_2)) \wedge \\
&((C(x,z) \wedge U(z,z_2)) \implies D(x,z_2))] \\
&\wedge \\
&[E(0,0) \wedge \\
&\neg(S(x,z) \wedge E(x,z)) \wedge \\
&\neg(S(x,z) \wedge S'(x,z)) \wedge \\
&\neg(E(x,z) \wedge S'(x,z)) \wedge \\
&(S(x+1,z) \implies S'(x,z)) \wedge \\
&(S'(x+1,z) \implies S'(x,z)) \wedge \\
&(E(z,x) \implies S(z,x+1)) \wedge \\
&(S(x,z) \iff E(x+1,z))] \\
&\wedge \\
&[Z(0,0) \wedge \\
&\neg Z(x+1,z) \wedge \\
&(Z(z,x) \implies Z(z,x+1))]
\end{aligned}
$$

7.2 Eliminating the Second Quantifier

For this subsection, we will be referring to rules/axioms in Statement 7.1. Eliminating the second quantifier z_3 takes more work and conceptual understanding. The statements we want to re-write are:

$$((A(z, z_2) \wedge B(x, z_2) \wedge A(z, z_3) \wedge B(x + 1, z_3)) \implies R(z_3, z_2))$$

and

$$((A(z, z_2) \wedge P(x, z_2) \wedge A(z, z_3) \wedge B(x, z_3)) \implies U(z_3, z_2))$$

It takes three quantifiers to reference a, b (z, x) and the respective column n (z_2). It takes a fourth quantifier (z_3) to reference the column containing a and $b + 1$.

We are not able to reference all such operands and columns with only three quantifiers (i.e. without z_3). (And similarly with a and p).

Therefore, instead, we must reference a and b components individually. The idea illustrated below is to create a new auxiliary predicate that acts as an equivalence relation on columns containing the same a value as the first operand. This enables us to reference the group (or rather, subset) of columns which have the same a value :

$$\tau(a) : \{n \in \mathbb{N} : A(a, n)\}$$

Furthermore, the idea we use to define V is to enforce the equivalence relation on any two natural numbers:

$$x \equiv_a z \iff (\exists a : x \in \tau(a) \wedge z \in \tau(a))$$

From the definition of τ and the understanding of the previous theorems, the equivalence relation does indeed exist. We define the predicate V, relative to the context of Statement 7.1 as follows:

$$\begin{aligned} ((A(x, z) \wedge A(x, z_2)) &\implies V(z, z_2)) \wedge \\ ((\neg A(x, z) \wedge A(x, z_2)) &\implies \neg V(z, z_2)) \wedge \\ ((A(x, z) \wedge \neg A(x, z_2)) &\implies \neg V(z, z_2)) \end{aligned}$$

Theorem 7.1. *V is an equivalence relation on $\mathbb{N}$. Furthermore $V(n, m)$ is true iff both columns reference the same 'a' operand (i.e. $A(a, n)$ and $A(a, m)$).*

Proof. Because each a (for every index column) is uniquely determined, comparing any two index column values n and m, either they have the same a operand, or they don't. If they don't then the last two implication consequences (implying $\neg V(z, z_2)$) hold. If they do, then the first implication consequence $V(z, z_2)$ holds. Hence, $V(n, m)$ can only be true iff they both store the same operand a. Hence, in other words $V(n, m) \iff (A(a, n) \iff B(a, m))$.

We now show V is an equivalence relation. To do this, we must show reflexivity, symmetry, and transitivity for any n:

$$(A(a, n) \wedge A(a, n)) \implies V(n, n)$$

Hence reflexivity holds.
To show symmetry, suppose $V(n, m)$ is true. We can take the contrapositive of the last two implications to conclude either $A(a, n)$ and $A(a, m)$ are both true or both false. Because there is some a such that they are both true, and by the reasoning described earlier, $V(m, n)$ must also be true, and symmetry holds.
To show transitivity holds, assume both $V(n, m)$ and $V(m, n')$ hold. Because the a operands at columns n and m match, and the a operands at columns m and n' match, then by proven conclusion, the a operands of n and n' match, and the implication that $V(n, n')$ is true follows.

□

Defining V in a way where it encodes an equivalence relation on $\tau(a)$ allows us to address the second operand b. Because the columns we are objectively trying to reference (i.e. store the column value of $f(a, b+1)$ in $f(a, b)$) are related by V, we can simply require $V(n, m)$ to be true (to fix a) to get the unique column referencing operands a and $b+1$. We can do this with the following implication:

$$((B(x, z) \wedge B(x+1, z_2) \wedge V(z, z_2)) \implies R(z_2, z)$$

We can apply this same logic to retrieve the row/column addressing the operands a and $p = a \times b$:

$$((P(x, z) \wedge B(x, z_2) \wedge V(z, z_2)) \implies U(z_2, z)$$

In summary, we eliminated the z_3 quantifier by expressing

$$((A(z, z_2) \wedge B(x, z_2) \wedge A(z, z_3) \wedge B(x+1, z_3)) \implies R(z_3, z_2)) \wedge \\ ((A(z, z_2) \wedge P(x, z_2) \wedge A(z, z_3) \wedge B(x, z_3)) \implies U(z_3, z_2))$$

with the following:

$$
\begin{aligned}
((A(x,z) \wedge A(x,z_2)) &\implies V(z,z_2)) \wedge \\
((\neg A(x,z) \wedge A(x,z_2)) &\implies \neg V(z,z_2)) \wedge \\
((A(x,z) \wedge \neg A(x,z_2)) &\implies \neg V(z,z_2)) \wedge \\
((B(x,z) \wedge B(x+1,z_2) \wedge V(z,z_2) &\implies R(z_2,z)) \wedge \\
((P(x,z) \wedge B(x,z_2) \wedge V(z,z_2) &\implies U(z_2,z))
\end{aligned}
$$

We can now re-write Statement 7.1 as follows:

Statement 7.2.

$$\forall x \forall z \forall z_2 \in \mathbb{N} :$$

$$
\begin{array}{r}
[A(0,0)\ \wedge \\
B(0,0)\ \wedge \\
C(0,0)\ \wedge \\
((A(z,x) \wedge A(z_2,x)) \implies E(z,z_2))\ \wedge \\
((B(z,x) \wedge B(z_2,x)) \implies E(z,z_2))\ \wedge \\
((C(z,x) \wedge C(z_2,x)) \implies E(z,z_2))\ \wedge \\
((C(x,z) \wedge S(z,z_2)) \implies (C(x,z_2) \vee C(x+1,z_2)))\ \wedge \\
((C(x,z) \wedge A(x,z) \wedge S(z,z_2)) \implies \neg A(x+1,z_2))\ \wedge \\
((\neg C(x,z) \wedge A(x,z) \wedge S(z,z_2)) \implies A(x+1,z_2))\ \wedge \\
((B(x+1,z) \wedge S(z,z_2)) \implies B(x,z_2))\ \wedge \\
((Z(z,x) \wedge B(z,x)) \implies A(z,x+1))\ \wedge \\
((Z(z_2,x) \wedge B(z_2,x) \wedge C(z,x)) \implies \neg C(z,x+1))\ \wedge \\
((Z(z_2,x) \wedge \neg B(z_2,x) \wedge C(z,x)) \implies C(z,x+1))\ \wedge \\
((Z(z_2,x) \wedge A(z_2,x) \wedge C(z,x)) \implies B(z,x))] \\
\wedge \\
[((D(z,x) \wedge D(z_2,x)) \implies E(z,z_2))\ \wedge \\
((P(z,x) \wedge P(z_2,x)) \implies E(z,z_2))\ \wedge \\
((R(z,x) \wedge R(z_2,x)) \implies E(z,z_2))\ \wedge \\
((U(z,x) \wedge U(z_2,x)) \implies E(z,z_2))\ \wedge \\
((Z(z,x) \wedge B(z,x)) \implies D(z,x))\ \wedge \\
((Z(z_2,x) \wedge B(z_2,x)) \implies \neg P(z,x))\ \wedge \\
((Z(z_2,x) \wedge B(z_2,x)) \implies \neg U(z,x))\ \wedge \\
((A(x,z) \wedge A(x,z_2)) \implies V(z,z_2))\ \wedge \\
((\neg A(x,z) \wedge A(x,z_2)) \implies \neg V(z,z_2))\ \wedge \\
((A(x,z) \wedge \neg A(x,z_2)) \implies \neg V(z,z_2))\ \wedge \\
((B(x,z) \wedge B(x+1,z_2) \wedge V(z,z_2)) \implies R(z_2,z))\ \wedge \\
((P(x,z) \wedge B(x,z_2) \wedge V(z,z_2)) \implies U(z_2,z))\ \wedge \\
((D(x,z) \wedge R(z_2,z)) \implies P(x,z_2))\ \wedge \\
((C(x,z) \wedge U(z,z_2)) \implies D(x,z_2))] \\
\wedge \\
[E(0,0)\ \wedge \\
\neg(S(x,z) \wedge E(x,z))\ \wedge \\
\neg(S(x,z) \wedge S'(x,z))\ \wedge \\
\neg(E(x,z) \wedge S'(x,z))\ \wedge \\
(S(x+1,z) \implies S'(x,z))\ \wedge \\
(S'(x+1,z) \implies S'(x,z))\ \wedge \\
(E(z,x) \implies S(z,x+1))\ \wedge \\
(S(x,z) \iff E(x+1,z))] \\
\wedge \\
[Z(0,0)\ \wedge \\
\neg Z(x+1,z)\ \wedge \\
(Z(z,x) \implies Z(z,x+1))]
\end{array}
$$

8 References

[TODO]